

Gradient Descent

first order optimization

Formulae:

$$x_{n+1} = x_n - (\text{learning rate}) \times \frac{dy}{dx}$$

Two ~~to~~ value given:-

$$\alpha = 0.01$$

Initial Guess value = 3

Q.1) $f(x) = x^2$

$$\frac{dy}{dx} = 2x$$

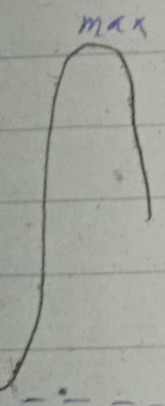
$$x_1 = x_0 - \alpha \left(\frac{dy}{dx} \right)$$

$$= 3 - (0.01)(2x)$$

$$= 3 - (0.01)(2(3))$$

$$= 3 - (0.01)(6)$$

$$\boxed{x_1 = 2.94}$$



Q.1)
Test
Pre
al

$$x_2 = x_1 - \alpha \left(\frac{dy}{dx} \right)$$

$$= 2.94 - (0.01) (2 \times 2.94)$$

$$\boxed{x_2 = 2.88}$$

$$x_3 = x_2 - \alpha \nabla f(x, y)$$

$$= 2.88 - (0.01) (2 \times 2.88)$$

$$\boxed{x_3 = 2.82}$$

$$x_4 = 2.82 - (0.01) (2 \times 2.82)$$

$$\boxed{x_4 = 2.76}$$

Q-2 $\sin^3(e^{x^2+2x})$

$$\alpha = 0.01$$

$$3 \sin^2 e^{x^2+2x} \cos(e^{x^2+2x}) (e^{x^2+2x}) (2x+2)$$

Q $\ln^5(\sin 2x)$

$$\frac{5 \ln^4(\sin 2x) (\cos 2x)}{\sin 2x}$$

Hessian Matrix $\rightarrow$ 2nd order
optimization

Formula:-

$$H \Rightarrow \begin{pmatrix} \frac{\partial^2 F}{\partial x^2} & \frac{\partial^2 F}{\partial x \partial y} \\ \frac{\partial^2 F}{\partial y \partial x} & \frac{\partial^2 F}{\partial y^2} \end{pmatrix} \text{ at } (1,1)$$

$$F(x,y) = x^2y + y^2x$$

$$\frac{\partial F}{\partial x} = 2xy + y^2$$

$$\frac{\partial^2 F}{\partial x \partial y} = 2x + 2y$$

$$F = x^2y + y^2x$$

$$\frac{\partial F}{\partial y} = x^2 + 2yx$$

$$\frac{\partial F}{\partial x \partial y} = 2x + 2y$$

$$\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix} \text{ at } (1,1)$$

$$Q) \quad f(x, y) = e^{y \ln x} \quad f(m, y) = e^{y \ln x}$$

$$f(x, y) = e^{y \ln x}$$

$$\frac{\partial f}{\partial x} = e^{y \ln x} \left(\frac{y}{x} \right)$$

$$\frac{\partial^2 f}{\partial x \partial y}$$

$$\frac{\partial f}{\partial y} = e^{y \ln x} (\ln x)$$

Bisection Method: Example

↳ is all about finding the root of an equation
which is of the form $f(x) = 0$

$$x^3 = 20$$

$$x_{\text{lower limit}} = 1 ; x_{\text{upper limit}} = 4$$

conduct 3 iterations

↳ initial guess

↳ initial guess

$$f(x) = x^3 - 20 = 0$$

• check whether
function is changing
sign

$$1) f(x_l) = f(1) = 1^3 - 20 = -19$$

$$f(x_u) = f(4) = 4^3 - 20 = 44$$

$$f(1) f(4) = -19 * 44 < 0$$

since, the function is changing sign b/w
1 and 4 then there should be
at least one root b/w 1 and 4

iteration 1

$$x_m = \frac{1 + 4}{2} = 2.5$$

↓
mid point

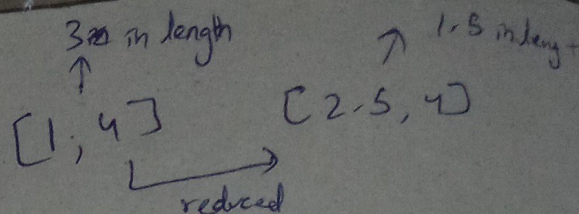
$$f(x_l) = f(1) = 1^3 - 20 = -19$$

$$f(x_m) = f(2.5) = 2.5^3 - 20 = -4.375$$

$$f(x_l) * f(x_m) = -19 * -4.375 > 0$$

$$\begin{matrix} \downarrow & \downarrow \\ 1 & 2.5 \end{matrix}$$

The root cannot be b/w 1 and 2.5, it has to be b/w 2.5 and 4
b/c that's the upper limit



New bracket = $[2.5, 4]$

iteration 2 $[2.5, 4]$

$$x_m = \frac{2.5 + 4}{2} = 3.25$$

$$f(x_l) = f(2.5) = -4.375$$

$$f(x_m) = f(3.25) = 14.328$$

$$f(x_l) f(x_m) = f(2.5) f(3.25) = (-4.375)(14.328) < 0$$

New bracket ~~$[2.5, 3.25]$~~ $[2.5, 3.25]$

calculating

Absolute relative approximate error -

$$|E_a| = \left| \frac{3.25 - 2.5}{3.25} \right| \times 100 \approx 23\%$$

$$\left| \frac{\text{new value} - \text{old value}}{\text{new value}} \right| \times 100$$

Iteration 3

$$x_m = \cancel{2.5} \frac{2.5 + \frac{3.25}{2}}{2} = \cancel{2.25} 2.875$$

$$f(x_l) = f(2.5) = -4.375$$

$$f(x_m) = f(2.875) = 3.763$$

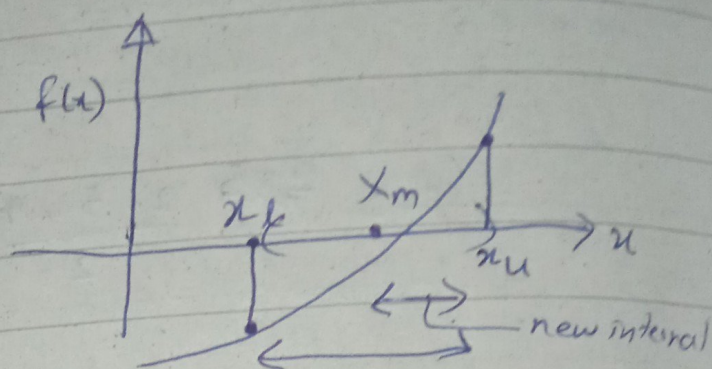
$$f(x_l) f(x_m) = f(2.5) f(2.875) \\ = (-4.375)(3.763) < 0$$

New bracket $[2.5, 2.875]$

if > 0 then change x_l and

Bisection Method

$$f(x) = 0$$



(1) choose x_l and x_u as the initial guesses
such that $f(x_l) f(x_u) < 0$

$$(2) \quad x_m = \frac{x_l + x_u}{2}$$

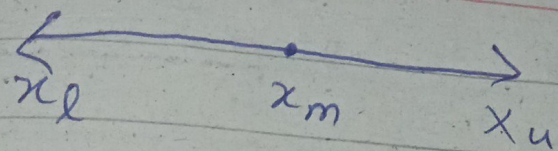
(3) Find $f(x_l) f(x_m)$ $\begin{cases} < 0 \\ > 0 \\ = 0 \end{cases}$

3 three possibilities

1) Either the function is changing sign b/w
these two points then $f(x_l) f(x_m) < 0$

2) Not changing sign > 0

(3) $f(x_l) f(x_m) = 0$ in that case
then $f(x_m)$ is root itself



~~if this be~~

1) if $f(x_l) f(x_m) < 0$

then $x_l = x_l$; $x_u = x_m$

2) if $f(x_l) f(x_m) > 0$

then $x_l = x_m$; $x_u = x_u$

3) if $f(x_l) f(x_m) = 0$

then root = x_m stop

4) Find $x_m = \frac{x_l + x_u}{2}$

Find $|\epsilon_a| = \left| \frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}} \right| \times 100$
 $\downarrow$
 relative approximate error

check if $|\epsilon_a| \leq \epsilon_s$
 $\downarrow$
 prespecified tolerance

⑤ go back to step 3

Lagrange Interpolation: Interval unequal & missing value find out

find the value of $y(10)$ when $y(5) = 12$,
 $y(6) = 13$, $y(9) = 14$, $y(11) = 16$

	x	y	
x_0	5	12	y_0
x_1	6	13	y_1
x_2	9	14	y_2
x_3	11	16	y_3

$$y(10) = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} y_0 +$$

$$\frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} y_1 +$$

$$\frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} y_2 +$$

$$\frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} y_3$$

$$\frac{(10-6)(10-9)(10-\overset{11}{13})(12)}{(5-6)(5-9)(5-13)} +$$

$$\boxed{\text{Second}} + \boxed{\text{Third}} + \boxed{\text{Fourth}}$$

$$\frac{(4)(1)(-\overset{-1}{8})(12)}{(-1)(-4)(-\overset{-6}{8})}$$

$$= 14.66$$

Differential Equation

$$\text{Euler method (formula)}: y_{n+1} = y_n + h f(x_n, y_n)$$

$$h = \frac{\text{final value} - \text{initial value}}{\text{step size}}$$

Q) $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y(0) = 1$

find y approximately for $x=0.1$
for Euler method.

Given

$y(0) = 1$ It means that $(x_0, y_0) = (0, 1)$

$$h = \frac{0.1 - 0}{5} = 0.02$$

Step 1 $y_{n+1} = y_n + h f(x_n, y_n)$

for $n = 0$

$$y_{0+1} = y_0 + h f(x_0, y_0)$$

$$= 1 + 0.02 \left(\frac{1-0}{1+0} \right) = 1.02$$

then

(x_0, y_0)

$$x_1 = x_0 + h = 0 + 0.02 = 0.02$$

Step 2

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for $n=1$

$$y_{1+1} = y_1 + h f(x_1, y_1)$$

$$y_2 = 1.02 + 0.02 \left(\frac{1.02 - 0.02}{1.02 + 0.02} \right)$$

$y_0 = (0, 1$

$$\boxed{y_2 = 1.039}$$

$$x_2 = x_1 + h$$

$$x_2 = 0.02 + 0.02 = 0.04$$

Now,

$$x_3, y_3 = ?$$

$$x_4, y_4 = ?$$

$$x_5, y_5 = ?$$

5 step size hence
find till x_5, y_5

-02

step 3

$$y_{n+1} = y_n + h f(x_n, y_n)$$

for $n = 2$

$$y_{2+1} = y_2 + h f(x_2, y_2)$$

$$y_3 = 1.039 + 0.02 \left(\frac{1.039 - 0.04}{1.039 + 0.04} \right)$$

$$y_3 = 1.057$$

$$x_3 = x_2 + h$$

$$= 0.04 + 0.02 = 0.06$$